

Mud Program Deviation: Management of Change & Approval Authority (illustrative)

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1. Objective

Defines formal governance and digital audit trail protocols when operational conditions dictate alteration of approved Well Operations Programs in deepwater blocks.

2. Thresholds Requiring Level 2 MOC

Any of the following conditions triggers mandatory Level 2 MOC approval:

- Mud density increase $> 0.2 \text{ ppg}$ above approved program.
- Planned entry into overpressured sands with narrow margin $< 1.0 \text{ ppg}$.
- Implementation of mechanical ROP caps during drilling breaks.

3. Required Approval Chain

1. **Initiator:** Drilling Superintendent / RTOC AI Co-pilot generates MOC memo with technical justification.
2. **Technical Endorsement:** Mud Chemist confirms barite inventory, hopper throughput, and rheological stability.
3. **Operational Endorsement:** Offshore Rig Company Man verifies surface pit room, trip tank integrity, and well-control readiness.
4. **Final Approval Authority:** Asset Lead / General Manager (Drilling), ONGC RTOC.

4. Digital Ledger & Compliance

All MOC requests, automated risk assessments, and signature hashes are recorded in the immutable RTOC ledger to satisfy Directorate General of Hydrocarbons (DGH) regulatory standards.